Precision in Algorithms

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Precision in Algorithms

> Prepare Precision in Algorithms: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Translate a short paper calculation into a Python script

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/1, ai-3/23, python-bridge/1. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Expo and Defense

Without opening the old lesson, explain one key idea from Expo and Defense. Give an example and a mistake that the idea helps you avoid.

Look for: Evaluate a saved candidate and a manual baseline fairly

Reference example, not the only acceptable wording: Require an identifiable saved version, complete instructions, valid whole minutes 0 to 30, Yes or No quiet **input**, inclusive time AND quiet compatibility, first-fit order, reset, stopping and clarification.

The baseline checks all rows and selects the first fit under the same contract. Accept honestly stated known failures; do not award a stronger result for hiding them.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Write an **algorithm** for borrowing a classroom book when the book may already be on loan. Include **input**s, a no-copy-available outcome and a check that prevents two borrowers receiving the same copy.

Concept to explain

Computers follow stated steps, not intended meaning.

Term	Meaning
algorithm	A specified sequence of steps for performing a task or solving a class of problems.
precise	Specific enough that a reader can distinguish the intended meaning or reproduce the action.
assumption	Something treated as true for reasoning that has not been established by the supplied evidence.
test case	A specified input and conditions with an expected result or measurement rule.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

This is a first encounter with saved Python scripts, not a compressed implementation of the Grade 6 adviser. Teach **assignment**, multiplication and print only as needed; values and types follow in Week 2.

Use a fresh run to avoid hidden interactive state. Reference: Python tutorial, An Informal Introduction to Python, https://docs.python.org/3/tutorial/introduction.html (**assignment**, arithmetic and print).

Readiness answer

Record four and six, multiply them to get 24, then display 24. The order must make the required values available before use.

Teacher-prepared Python 3 editor with a new saved .py file and a visible **output** area; test Run before class

Paper counters or equal-group sketches and a blank expected/observed record

The printed script and workbook; no logins or personal data required

End goal for the pictorial case

By the end, I can translate an ordered **algorithm** into Python and check state after each line.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Precision in Algorithms using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Vague: ‘Add the filling.’

2. Question: Which filling? How much? Where?

3. **Precise**: place 2 cucumber slices on the center of bread A.

4. Test with a new partner: identical results are **evidence** of precision.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

A sandwich **algorithm** says add spread, close it, cut it, without stating materials or directions.

The teaching case

Expose **assumption**s and write a **precise** paper simulation.

Exact explanation to model

Why the result follows

Use two paper bread squares, mark a spoon-sized spread area on one, place the second on top and draw a diagonal cut. State materials and order. No real food or sharp tools are needed.

Explain the added worked example

Explain the added worked example · 1

In Grade 6 you traced rules on paper. Now you will write a small Python **program** and check what it actually does. Begin with a counter-packing calculation, not the whole activity adviser.

The requirement is to find the total counters in equal packs and display that total. For this first **program**, the supplied quantities are whole, nonnegative numbers; it does not yet validate typed **input**.

Explain the added worked example · 2

Use a teacher-prepared Python 3 editor that can run a saved .py file. Open a new file called counters.py, type the code exactly as shown in the next section, save it, and use the editor’s Run command.

Read the **output** area separately from the source editor. No account, AI tool, package installation or personal data is needed. If a device is unavailable, trace on paper and label the result predicted rather than executed.

Explain the added worked example · 3

Read the script from top to bottom. The equals sign assigns the value on its right to the name on its left. The star means multiplication. print(total) displays the value of total; it does not display the letters in the name.

Type ordinary straight characters and keep the parentheses. The Python prompt symbols >>> are not part of this saved script.

Key terms

Term	Meaning
Algorithm	An ordered procedure for solving a stated problem
Program	Instructions written in a language a computer can execute
Interpreter	Software that executes Python instructions
Assignment	Binding a name to the value of an expression
Output	Information a program displays or returns

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

After line	Names and values	Displayed output
packs = 4	packs is 4	Nothing yet
per_pack = 6	packs is 4; per_pack is 6	Nothing yet
total = packs * per_pack	total is 24	Nothing yet
print(total)	Values are unchanged	24

Model the reasoning aloud · 4

Write 24 as the expected **output** before running. Then copy the actual **output** into a separate column. A matching result supports this case, not every possible **input**.

A saved file may contain an old edit: save again and rerun the whole script. Do not type the expected number into the **output** record without running.

> **Why is there no displayed output after the third line? What changes if print(total) is placed first in a fresh run, before any assignment? Explain using which names exist.**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Precision in Algorithms

A fictional book-box counter has three boxes with five books each and two loose books.

The facts we will use

> **Supplied facts**
>
> Python instructions: boxes=3; total=boxes*5; total=total+2; print(total).

> **Supplied facts**
>
> Run the instructions in this order with ordinary integer arithmetic.

AI ACADEMY / AI-2/1

Precision in Algorithms

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Read input

boxes is assigned 3.



2. Multiply

total becomes 3 x 5=15.



3. Update total

total becomes its current value 15 plus 2: 17.



4. Print

The program prints 17 once.

CHANGE ONE THING

Set boxes=0, keeping the two loose books.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read input	boxes is assigned 3.
2. Multiply	total becomes 3×5=15.
3. Update total	total becomes its current value 15 plus 2: 17.
4. Print	The program prints 17 once.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Assignment replaces a variable's stored value. In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left.

The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal. Predict **output** before running.

What this example does not establish

Keep the stated limit

These statements count supplied books; they do not validate **input** types or infer stock from measurements.

Change one thing in the pictured case

Change one thing

Change: Set boxes=0, keeping the two loose books.

Predicted result

Prediction: The printed result is 2.

Why the result follows

Reason: 0×5+2=2; there is no rule saying zero boxes means zero loose books.

Ask while showing the picture

Ask while showing the picture · 5

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Precision in Algorithms, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: A sandwich **algorithm** says add spread, close it, cut it, without stating materials or directions.

2. State the decision to explain

Write the question to explain: Expose **assumption**s and write a **precise** paper simulation.

3. Reveal the worked reasoning

Reveal after **prediction**s: Use two paper bread squares, mark a spoon-sized spread area on one, place the second on top and draw a diagonal cut. State materials and order. No real food or sharp tools are needed.

Hold back the independent task

Keep for an independent attempt: A robot starts at square 0. Replace move a bit and stop nearby with instructions that put it exactly on square 3 using steps of one square.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

The third line assigns total but does not call print, so nothing is displayed. A first-line print(total) raises NameError in a fresh run because total has not been assigned.

Exit: 11 is wrong for five groups of six; addition is a likely cause, and multiplication should give 30. Retain the old result and check the repaired file with both pack counts. Record whether the work was executed or traced.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve equal-group multiplication and collect the prerequisite explanation.
7 to 20 minutes	Model source versus output, save counters.py and translate each paper step with the diagram.

Stage	Teacher action
20 to 33 minutes	Complete W1 and predict the trace before running the four-line program.
33 to 45 minutes	Start W2 and distinguish observation from the expected answer; record device access.
Session 2 start to 8 minutes	Rerun with five packs and compare the two records.
8 to 20 minutes	Use W3 and the guided question to diagnose a first-line print in a fresh run.
20 to 35 minutes	Collect W4 independently before hints; observe whether each pupil can explain zero trays.
35 to 45 minutes	Complete W5 and the exit. Save source and records; identify pupils needing editor support.

Reduce help deliberately

Completion practice

Completion task: Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

Reasoning to listen for

Teacher check: Expected 24 then 30. Accept actual errors with an accurate account; do not replace **observation**s. Record the changed pack count, saved file and run or trace status.

Two matching cases do not show all **input**s are handled.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **Expected repair**
>
> Expected correction: **Assignment** binds a name; print produces the displayed **output**.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

> If print(total) moves before total=total+2, what gets printed for the original **input**?

Reasoned teacher answer

It prints 15. The later update still makes total 17, but nothing then prints the new value.

Changed-case reasoning: The printed result is 2. 0×5+2=2; there is no rule saying zero boxes means zero loose books.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Precision in Algorithms investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Trade sandwich instructions, follow them literally, mark every place where two interpretations are possible, then repair one cause at a time.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

An equals sign displays the answer: **Assignment** binds a name; print produces the displayed **output**.

A successful run means the requirement was met: Compare the **output** with the paper rule and expected result.

Python guesses the intended order: A name must be assigned before this script reads it.

Typing a **program** trains an AI model: These statements apply an explicit multiplication rule.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or **assumption** and asking them to defend the effect, not simply by assigning more of the same questions.

TEACHER GUIDE · 5 OF 9

Assess and explain the answers

> Use the Precision in Algorithms questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

Use two paper bread squares, mark a spoon-sized spread area on one, place the second on top and draw a diagonal cut. State materials and order. No real food or sharp tools are needed.

Independent question

Independent question · withhold the key

A robot starts at square 0. Replace move a bit and stop nearby with instructions that put it exactly on square 3 using steps of one square.

Expected independent answer

Teacher key · Expected independent answer

Move forward one square three times, then stop. The distance, unit, repetition count and stopping point are specified.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A human's guess makes an ambiguous **algorithm precise**.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

> #### Teacher key · Original transfer answer
>
> Read book-copy ID and availability; if unavailable, report no copy; otherwise record the loan and mark the copy unavailable before confirming. A second request sees unavailable.
>
> A paper simulation should test available, already borrowed and unknown-copy cases without real student identities.

Explanations for the additional workbook examples

W1 Connect the paper steps to Python

Workbook question · W1

Write the complete four-line script for 4 packs of 6. Beside each line explain what it does. Which line displays information?

Teacher reasoning · W1

packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

W2 Predict then observe

Workbook question · W2

Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

Teacher reasoning · W2

Expected 24 then 30. Accept actual errors with an accurate account; do not replace **observation**s. Record the changed pack count, saved file and run or trace status. Two matching cases do not show all **input**s are handled.

W3 Explain an order error

Workbook question · W3

Move print(total) to the first line of a fresh script. Predict what happens and why. Put the lines back into an order that meets the requirement.

Teacher reasoning · W3

The first line tries to read total before it exists, causing NameError in a fresh run. Assign packs and per_pack, calculate total, then print it.

Earlier values lingering in an interactive session can hide the problem; run the complete saved script afresh.

W4 Transfer the procedure independently

Workbook question · W4

A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again. Explain why adding the two quantities is wrong.

Teacher reasoning · W4

trays = 3; per_tray = 7; tiles = trays * per_tray; print(tiles), on separate lines. **Output**s 21 and then 0. Zero trays contain zero tiles. Addition would give 10 then 7, not the number in equal groups.

W5 Keep an honest conclusion

Workbook question · W5

Explain the difference between a successful run and a correct calculation. State one limit of this **program** and what you would keep so another person can check your work.

Teacher reasoning · W5

A successful run means the execution finished without a reported error; correctness requires matching the stated total-count rule. The script assumes supplied whole nonnegative quantities and does not validate arbitrary **input**.

Keep source, quantities, expected and actual **output** and whether it was run or traced.

Assess independent transfer

Independent question · withhold the key

Selected case: A fictional shelf has 3 trays of 7 tiles each. Write a four-line Python script using trays, per_tray and tiles. Predict its **output**. Then change only trays to 0 and predict again.

Explain why adding the two quantities is wrong.

Selected case answer

Teacher key · Selected case answer

trays = 3; per_tray = 7; tiles = trays * per_tray; print(tiles), on separate lines. **Output**s 21 and then 0. Zero trays contain zero tiles. Addition would give 10 then 7, not the number in equal groups.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Precision in Algorithms to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Write exact scoring steps so two people rank the same book the same way.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 2: Decompose a Goal. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Offer four separate line cards, a source/**output** label and a multiplication sketch. Allow a paper trace when devices are unavailable, with execution practice still pending.

Do not supply the independent transfer numbers' answers. Extension: explain why a plus-sign version can accidentally agree with multiplication for two packs of two, and why one passing example is weak **evidence**.

W4 must give a complete ordered script and explain **output**s 21 and 0 using equal groups. Distinguish an independent paper trace from successful use of the editor; record both skills separately.

After support, retry with 2 boxes of 9 pencils, then 0 boxes: 18 and 0. Ask for the print line and why it follows the calculation.

Week 2 studies values, variable names and types, including the difference between numeric data and text.

Plan the delayed return

Suggested return points in this level: ai-2/2, ai-2/4, ai-2/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Use small predicted traces when translating the Transparent Book Recommender's paper logic into code.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Precision in Algorithms lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

> Teaching target: Translate a short paper calculation into a Python script

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Make records for 4 packs of 6, then 5 packs of 6. Predict before running. Record actual **output**, any error and what changed. If working on paper, label both results as traces.

Supported practice answer

Reasoned practice answer: Expected 24 then 30. Accept actual errors with an accurate account; do not replace **observation**s. Record the changed pack count, saved file and run or trace status.

Two matching cases do not show all **input**s are handled.

Check every learner before discussion: Use the worked case for From a paper **algorithm** to Python. Proposed explanation: “An equals sign displays the answer”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: **Assignment** binds a name; print produces the displayed **output**.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Set boxes=0, keeping the two loose books. Require a **prediction** and a reason before comparing. Expected result: The printed result is 2. Reason: 0×5+2=2; there is no rule saying zero boxes means zero loose books.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

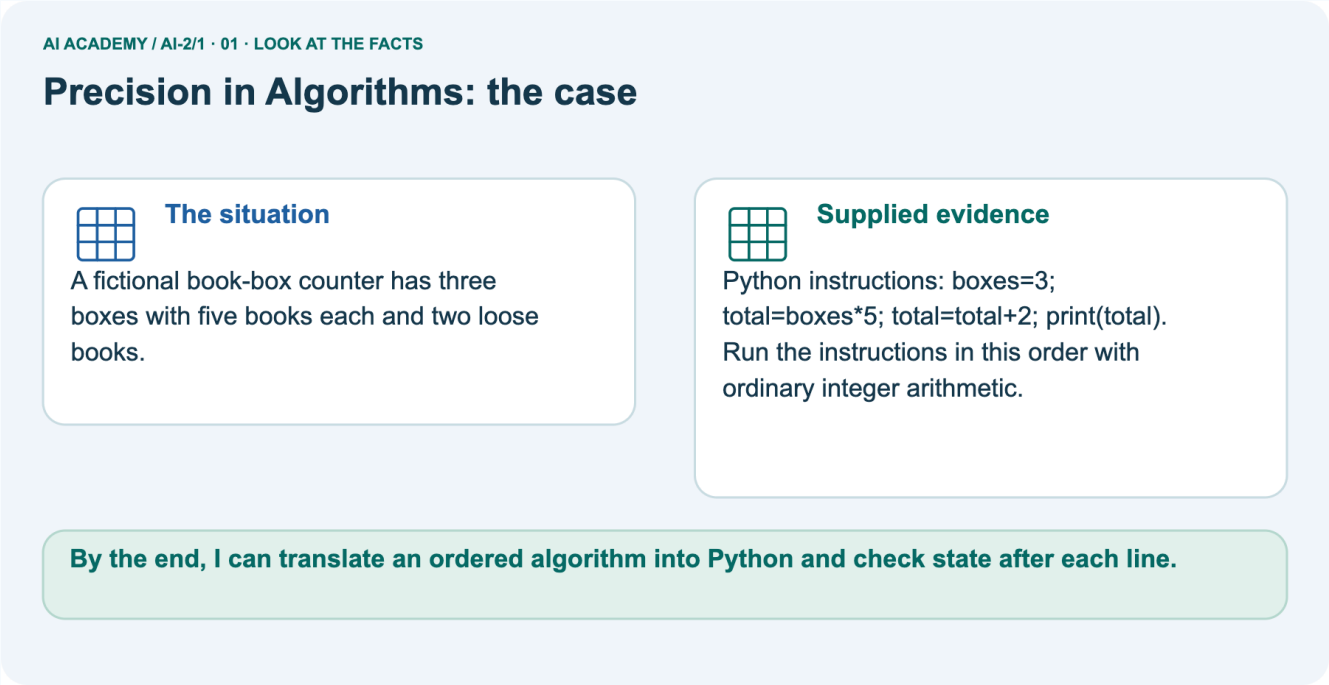
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Precision in Algorithms graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Precision in Algorithms: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-box counter has three boxes with five books each and two loose books. Supplied facts: Python instructions: boxes=3; total=boxes*5; total=total+2; print(total).

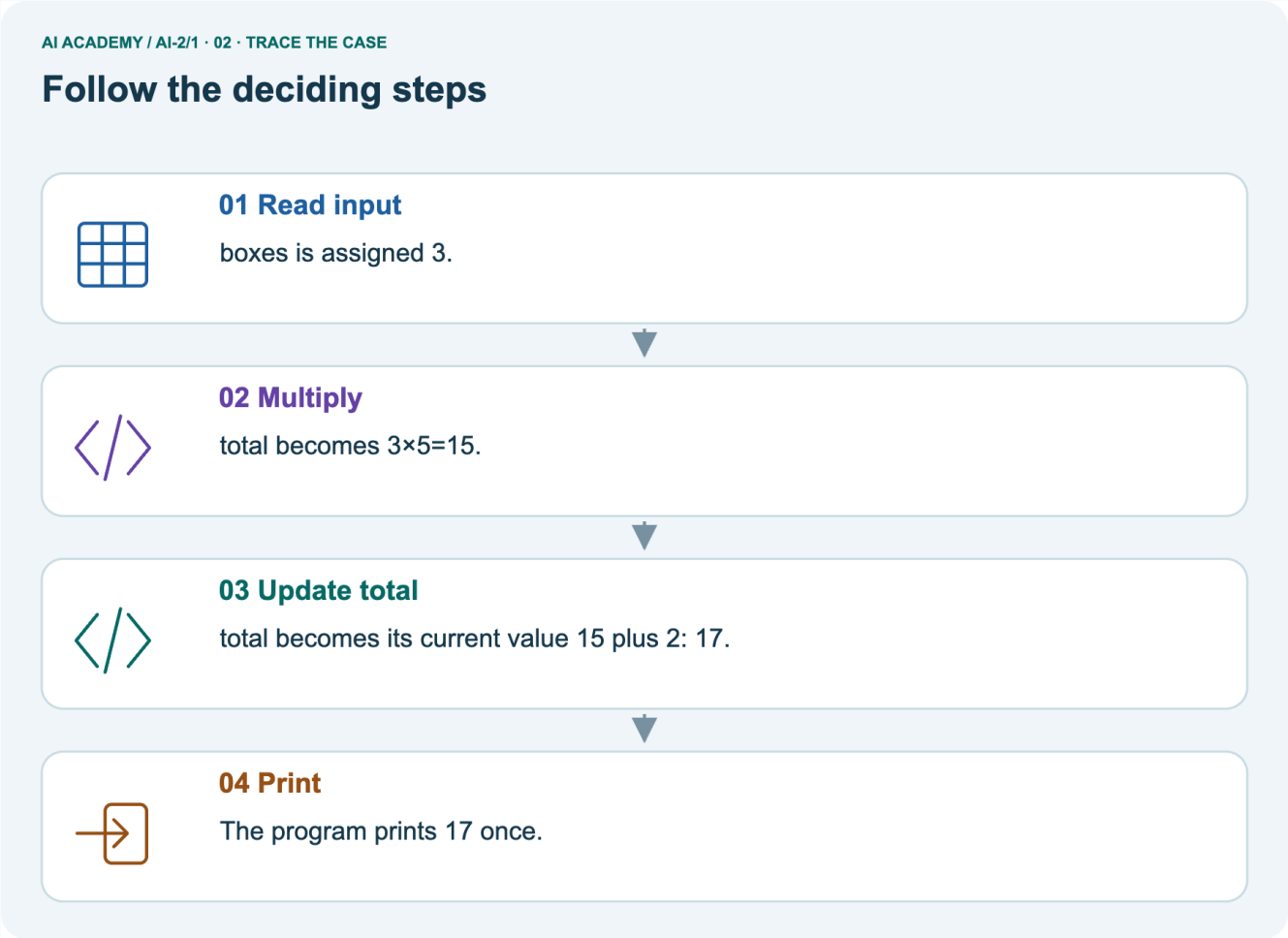
Run the instructions in this order with ordinary integer arithmetic.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read **input**: boxes is assigned 3. → Multiply: total becomes 3×5=15. → Update total: total becomes its current value 15 plus 2: 17. → Print: The **program** prints 17 once.

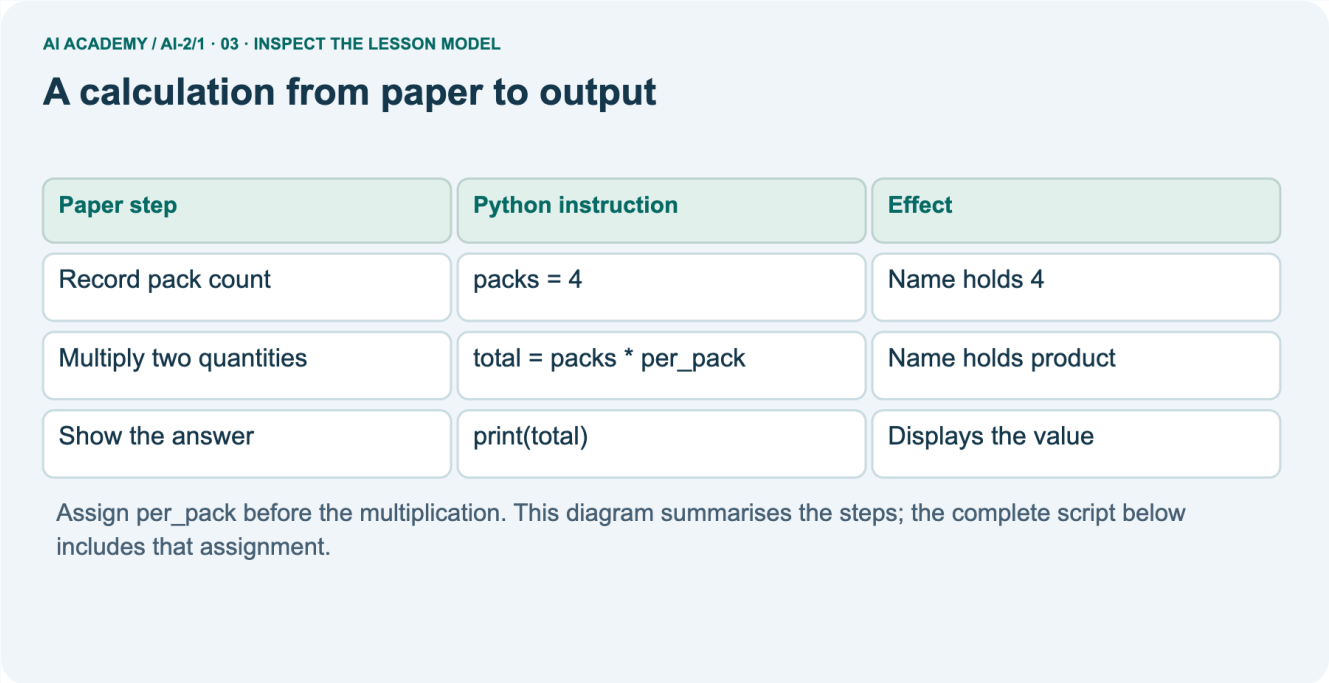
Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if

the image is difficult to see.

03. A calculation from paper to output



Follow the labelled picture with the matching explanation. Open the vector diagram

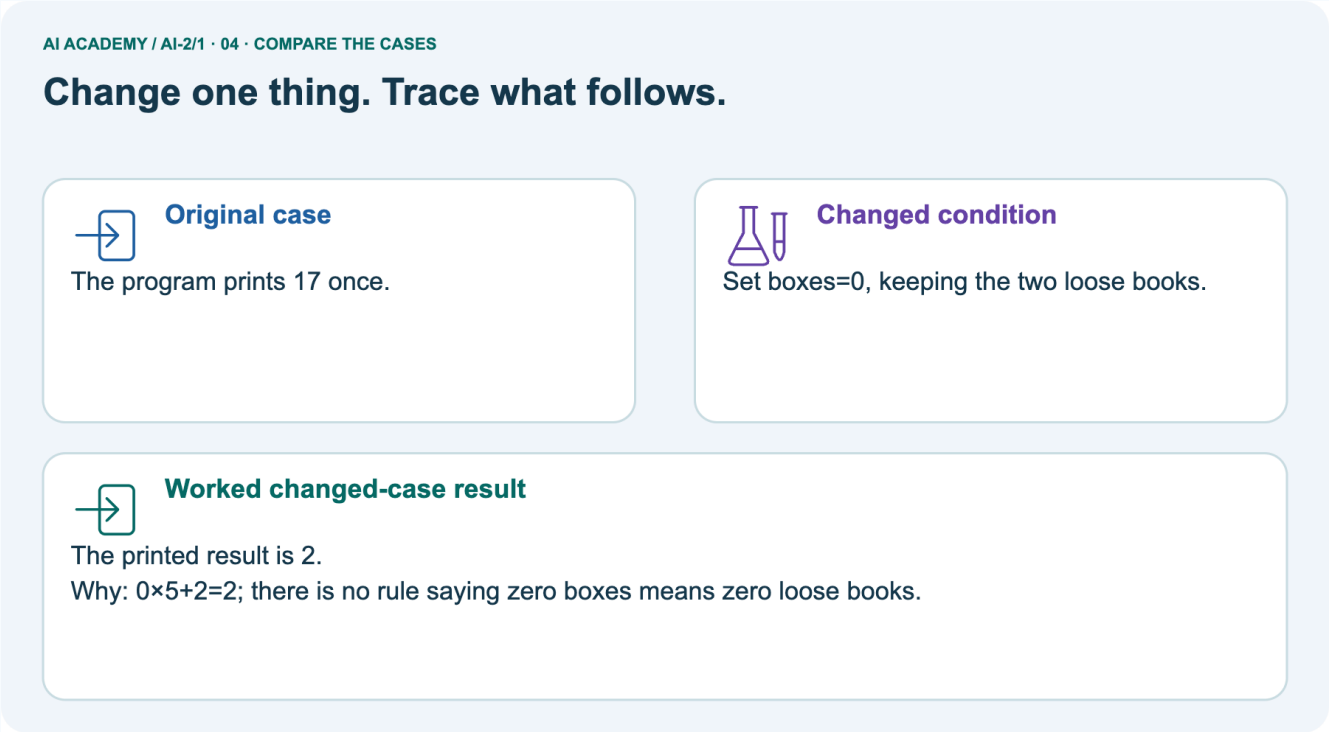
A calculation from paper to **output**. Record pack count | packs = 4 | Name holds 4. Multiply two quantities | total = packs * per_pack | Name holds product. Show the answer | print(total) | Displays the value

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The **program** prints 17 once. Changed condition: Set boxes=0, keeping the two loose books. Expected result: The printed result is 2. Reason: 0×5+2=2; there is no rule saying zero boxes means zero loose books.

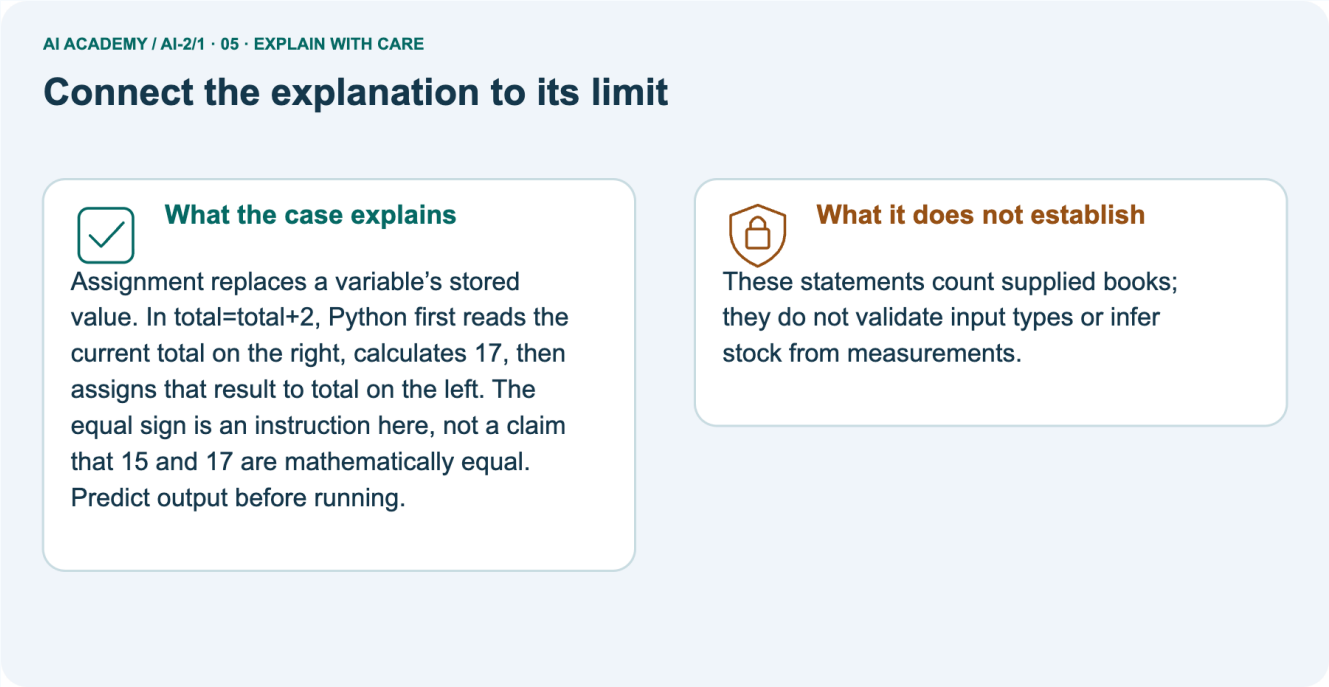
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 0×5+2=2; there is no rule saying zero boxes means zero loose books.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Assignment replaces a variable's stored value. In total=total+2, Python first reads the current total on the right, calculates 17, then assigns that result to total on the left.

The equal sign is an instruction here, not a claim that 15 and 17 are mathematically equal. Predict **output** before running.

Limit: These statements count supplied books; they do not validate **input** types or infer stock from measurements.

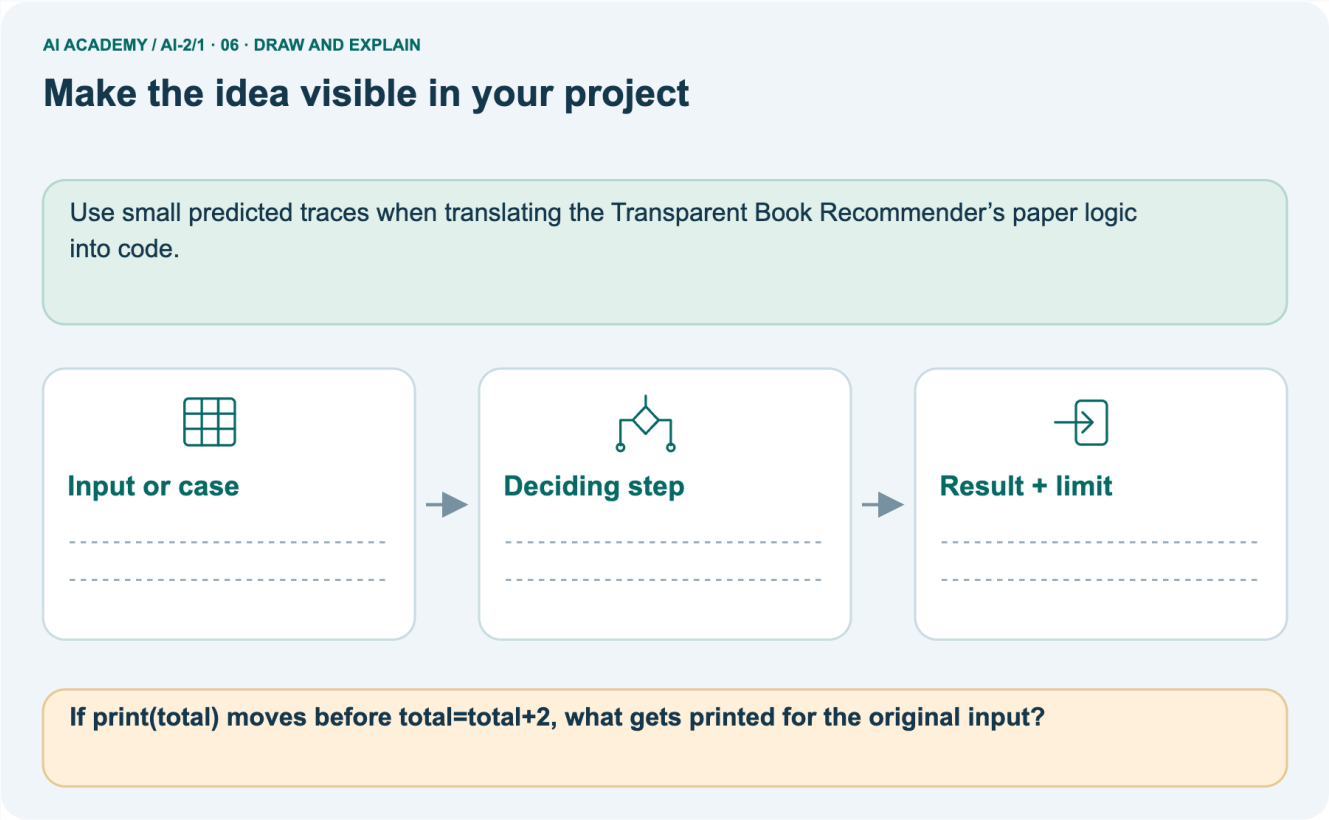
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These statements count supplied books; they do not validate **input** types or infer stock from measurements.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use small predicted traces when translating the Transparent Book Recommender's paper logic into code. Draw your **input**, deciding step, result and limit.

Supported question: If print(total) moves before total=total+2, what gets printed for the original **input**?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: It prints 15. The later update still makes total 17, but nothing then prints the new value.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.